SIMATS ENGINEERING

ASSESSMENT TOOL 2 (LOW) - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO2: R Programming Structures, Different types of loops such as: for (), while() loops, Control Statements, Looping Over Non-vector Sets,- If-Else, Arithmetic and Boolean Operators and values, Default Values for Argument, Return Values, Deciding Whether to explicitly call return- Returning Complex Objects, Functions are Objective, No Pointers in R, Recursion

(Bloom Level -BL4- BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

I S. V. K. Sharon Surya (192324246) *(Name / Reg No) certify that this submission is my original work and*
that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS: **Total Marks: 100**

Scenario

A retail company operates stores across multiple cities. The management has collected customer transaction data containing:

- Customer_ID
- Gender
- Age
- City
- Product_Category
- Purchase_Amount
- Monthly_Income
- Customer_Rating
- Membership_Status
- Store_Type

The company wants to improve customer satisfaction, optimize marketing strategies, and increase sales using data-driven decisions.

Question 1

The management wants to understand the overall spending behavior of customers. Calculate and interpret appropriate measures of central tendency (mean, median, and mode) for Purchase Amount. Based on the results, recommend which measure best represents customer spending patterns.

Question 2

Analyze the variability in customer spending using measures of dispersion such as range, variance, standard deviation, and interquartile range. Identify whether spending behavior is consistent across customers and justify your findings.

Question 3

The marketing team suspects that customer ratings differ significantly between premium and regular members. Perform an exploratory analysis of Customer Rating and summarize the key patterns using suitable statistical measures and visualizations.

Question 4

Investigate the relationship between Monthly Income and Purchase Amount using bivariate analysis techniques. Interpret the strength and direction of the relationship and discuss its implications for targeted marketing campaigns.

Question 5

The company wants to determine whether male and female customers differ in their average purchase amount. Formulate appropriate hypotheses, perform a T-test, and interpret the results to support business decision-making.

Question 6

Management wishes to compare customer satisfaction ratings across three different store types (Urban, Semi-Urban, and Rural). Conduct an ANOVA test and explain whether significant differences exist among the store categories.

Question 7

Analyze whether Membership Status and Product Category preference are associated. Perform a Chi-square test, interpret the findings, and explain how the results can support customer retention strategies.

Question 8

The company plans to launch a new loyalty program. Using EDA techniques, identify the customer segment that should be targeted first. Support your recommendation using appropriate statistical evidence.

Question 9

Based on customer transaction data, identify key factors influencing high purchase amounts. Utilize central tendency measures, dispersion measures, and bivariate analysis to justify your conclusions.

Question 10

Prepare a business analytics report summarizing the insights obtained from EDA, T-test, ANOVA, and Chi-square analysis. Provide actionable recommendations that can help management improve customer engagement and profitability.

ANSWERS:

1. data=read.csv("C:/Users/Dell/Downloads/customer_data.csv")
purchase <- data$Purchase_Amount
mean_value <- mean(purchase, na.rm = TRUE)
median_value <- median(purchase, na.rm = TRUE)
mode_value <- names(sort(table(purchase), decreasing = TRUE))[1]
cat("Mean:", mean_value, "\n")
cat("Median:", median_value, "\n")
cat("Mode:", mode_value, "\n")

OUTPUT:
> cat("Mean:", mean_value, "\n")
Mean: 6197.196
> cat("Median:", median_value, "\n")
Median: 5368.56
> cat("Mode:", mode_value, "\n")
Mode: 843.73

2. purchase <- data$Purchase_Amount
range_value <- max(purchase, na.rm = TRUE) - min(purchase, na.rm = TRUE)
variance_value <- var(purchase, na.rm = TRUE)
sd_value <- sd(purchase, na.rm = TRUE)
Q1 <- quantile(purchase, 0.25, na.rm = TRUE)
Q3 <- quantile(purchase, 0.75, na.rm = TRUE)
IQR_value <- Q3 - Q1
cat("Range:", range_value, "\n")
cat("Variance:", variance_value, "\n")
cat("Standard Deviation:", sd_value, "\n")
cat("IQR:", IQR_value, "\n")

OUTPUT:
> cat("Range:", range_value, "\n")
Range: 19671.85
> cat("Variance:", variance_value, "\n")
Variance: 13217692
> cat("Standard Deviation:", sd_value, "\n")
Standard Deviation: 3635.614
> cat("IQR:", IQR_value, "\n")
IQR: 5398.7

3. print(tapply(data$Customer_Rating,

```
             data$Membership_Status,
             mean,
             na.rm = TRUE))
print(tapply(data$Customer_Rating,
             data$Membership_Status,
             median,
             na.rm = TRUE))
print(tapply(data$Customer_Rating,
             data$Membership_Status,
             sd,
             na.rm = TRUE))
boxplot(Customer_Rating ~ Membership_Status,
        data = data,
        main = "Customer Rating by Membership Status",
        xlab = "Membership Status",
        ylab = "Customer Rating")
```

OUTPUT:

```
 Premium  Regular
3.852941 3.871429
Premium Regular
    3.9     4.0
Premium   Regular
0.7105921 0.7268654
```

4. correlation <- cor(data$Monthly_Income,

```
                   data$Purchase_Amount,
                   use = "complete.obs")
cat("Correlation:", correlation, "\n")
plot(data$Monthly_Income,
     data$Purchase_Amount,
     main = "Monthly Income vs Purchase Amount",
     xlab = "Monthly Income",
     ylab = "Purchase Amount",
     pch = 19)
abline(lm(Purchase_Amount ~ Monthly_Income, data = data))
```

OUTPUT:

Correlation: 0.7040942

```
5. male <- data$Purchase_Amount[data$Gender == "Male"]
female <- data$Purchase_Amount[data$Gender == "Female"]
result <- t.test(male, female)
print(result)
```

OUTPUT:

```
Welch Two Sample t-test

data:  male and female
t = -0.69358, df = 85.719, p-value = 0.4898
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1980.4146   955.9778
sample estimates:
mean of x mean of y
 5951.331  6463.549
```

```
6. model <- aov(Customer_Rating ~ Store_Type, data = data)
summary(model)
boxplot(Customer_Rating ~ Store_Type,
        data = data,
        main = "Customer Rating by Store Type",
        xlab = "Store Type",
        ylab = "Customer Rating")
```

OUTPUT:

```
 Df Sum Sq Mean Sq F value Pr(>F)
Store_Type  2  3.03  1.5171  3.093 0.0499 *
Residuals   97  47.58  0.4905
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
7. table_data <- table(data$Membership_Status,
                    data$Product_Category)
print(table_data)
result <- chisq.test(table_data)
print(result)
if(result$p.value < 0.05) {
  print("Reject H0: Significant association exists.")
} else {
  print("Fail to reject H0: No significant association exists.")
}
```

OUTPUT:
```
 Beauty Clothing Electronics Furniture Groceries
  Premium     11      10          6        9       15
  Regular     17      12          5        9        6

        Pearson's Chi-squared test

data:  table_data
X-squared = 5.3777, df = 4, p-value = 0.2507
"Fail to reject H0: No significant association exists."
```

8.
```
segment <- aggregate(
  cbind(Purchase_Amount, Monthly_Income, Customer_Rating)
  ~ Membership_Status,
  data = data,
  FUN = mean
)
print(segment)
boxplot(Purchase_Amount ~ Membership_Status,
        data = data,
        main = "Purchase Amount by Membership",
        xlab = "Membership Status",
        ylab = "Purchase Amount")
```

OUTPUT:
```
Membership_Status Purchase_Amount Monthly_Income Customer_Rating
1         Premium        6579.433       68184.16        3.852941
2         Regular        5799.358       62532.10        3.871429
```

9.
```
print(aggregate(Purchase_Amount ~ Membership_Status,
                data = data,
                FUN = mean))
print(aggregate(Purchase_Amount ~ Product_Category,
                data = data,
                FUN = mean))
print(aggregate(Purchase_Amount ~ Store_Type,
                data = data,
                FUN = mean))
cor(data[, c("Age",
             "Monthly_Income",
             "Purchase_Amount",
             "Customer_Rating")],
    use = "complete.obs")
plot(data$Monthly_Income,
```

```
     data$Purchase_Amount,
     main = "Income vs Purchase Amount",
     xlab = "Monthly Income",
     ylab = "Purchase Amount",
     pch = 19)
```

OUTPUT:

```
Membership_Status Purchase_Amount
1          Premium        6579.433
2          Regular        5799.358
 Product_Category Purchase_Amount
1          Beauty        4534.539
2        Clothing        5858.777
3     Electronics        6546.943
4       Furniture        9737.268
5       Groceries        5551.058
 Store_Type Purchase_Amount
1      Rural        6098.584
2 Semi-Urban        6509.757
3      Urban        6001.633
Age Monthly_Income Purchase_Amount Customer_Rating
Age             1.00000000     0.05527302     -0.01650482      0.04054636
Monthly_Income   0.05527302     1.00000000      0.70409425     -0.02059682
Purchase_Amount -0.01650482     0.70409425      1.00000000      0.15061932
Customer_Rating  0.04054636    -0.02059682      0.15061932      1.00000000
```

```
purchase <- data$Purchase_Amount
cat("Mean:", mean(purchase, na.rm = TRUE), "\n")
cat("Median:", median(purchase, na.rm = TRUE), "\n")
cat("Mode:", names(sort(table(purchase),
                        decreasing = TRUE))[1], "\n")
cat("Range:", diff(range(purchase, na.rm = TRUE)), "\n")
cat("Variance:", var(purchase, na.rm = TRUE), "\n")
cat("Standard Deviation:", sd(purchase, na.rm = TRUE), "\n")
cat("IQR:", IQR(purchase, na.rm = TRUE), "\n")
correlation <- cor(data$Monthly_Income,
                   data$Purchase_Amount,
                   use = "complete.obs")
cat("Income-Purchase Correlation:",
    correlation, "\n")
male <- data$Purchase_Amount[data$Gender == "Male"]
female <- data$Purchase_Amount[data$Gender == "Female"]
t_result <- t.test(male, female)
cat("T-Test p-value:",
```

```r
    t_result$p.value, "\n")
anova_model <- aov(Customer_Rating ~ Store_Type,
                   data = data)
anova_result <- summary(anova_model)
print(anova_result)
table_data <- table(data$Membership_Status,
                    data$Product_Category)
chi_result <- chisq.test(table_data)
cat("Chi-Square p-value:",
    chi_result$p.value, "\n")
```

OUTPUT:

```
Mean: 6197.196
Median: 5368.56
Mode: 843.73
Range: 19671.85
Variance: 13217692
Standard Deviation: 3635.614
IQR: 5398.7
Income-Purchase Correlation: 0.7040942
            Df Sum Sq Mean Sq F value Pr(>F)
Store_Type   2   3.03  1.5171   3.093 0.0499 *
Residuals   97  47.58  0.4905
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
T-Test p-value: 0.489824
Chi-Square p-value: 0.2506875
```

RUBRICS FOR CALCULATION

Industry Problem Solving Task					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Application of EDA Techniques	25	Correctly applies central tendency, dispersion, and bivariate analysis techniques.	Applies most EDA techniques correctly with minor omissions.	Applies basic EDA techniques with limited accuracy.	Demonstrates limited or incorrect application of EDA techniques.
Statistical Analysis & Testing	25	Correctly performs and interprets T-test, ANOVA, and Chi-square analysis.	Performs statistical tests with minor interpretation errors.	Performs basic statistical analysis with limited interpretation.	Demonstrates poor understanding of statistical tests.
Problem Analysis & Data Interpretation	20	Accurately analyzes data patterns and derives meaningful insights.	Provides reasonable analysis with adequate interpretation.	Provides limited analysis and interpretation.	Analysis is weak or incorrect.
Decision-Making & Business Recommendations	20	Provides data-driven recommendations that effectively address the business problem.	Recommendations are relevant and supported by analysis.	Recommendations are basic with limited justification.	Recommendations are weak, unsupported, or missing.
Organization & Completeness	10	Response is well-structured, complete, and addresses all requirements.	Mostly complete and organized.	Partially complete with some missing components.	Incomplete and poorly organized.